18/05/26

ARITHMETIC PROGRESSION

2, 4, 8, 16, 32, 64, ...

{1, 4, 7, 10, 13, ...} Arithmetic Progression

+3 (+3 +3 +3 +3) — Common difference

1, 2, 4, 7, 11, 16, ...

1, 1, 2, 6, 24, ...

2, 2, 2, 2, 2, 2, ... (+0 +0 +0 −)

Common difference = 0

ARITHMETIC PROGRESSION (AP).

An **arithmetic progression** is a list of numbers in which each term is obtained by adding a fixed number to the preceding term except the first term.

This fixed number is called the common difference of the AP. It can be positive, negative or zero.

Let us denote the first term of an AP by a_1, second term by a_2, . . ., nth term by a_n and the common difference by d. Then the AP becomes $a_1, a_2, a_3, \ldots, a_n$.

So, $a_2 - a_1 = a_3 - a_2 = \ldots = a_n - a_{n-1} = d$.

The general form of an arithmetic progression is given by

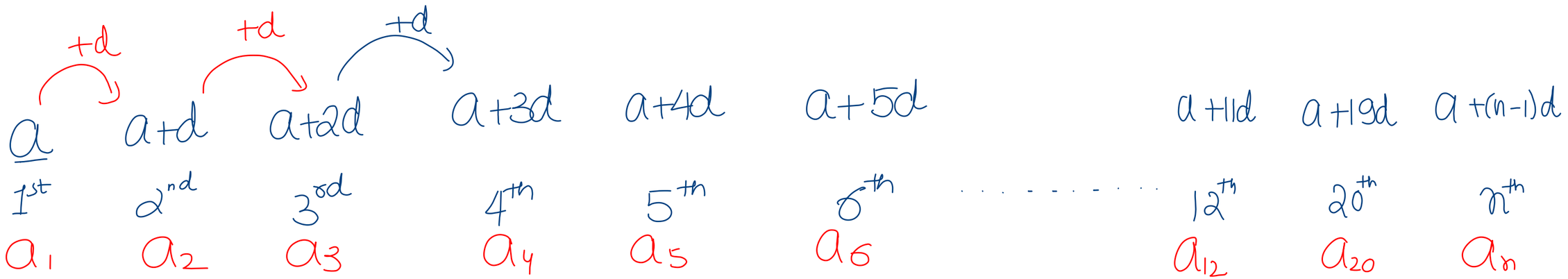
$$a, a + d, a + 2d, a + 3d, \ldots$$

where a is the first term and d the common difference.

10, 9, 8, 7, ...

N^{th} Term of AP

Let the first term of A.P is a & common difference is denoted by d.



$$a_n = a + (n-1)d$$

$a_n \rightarrow n^{th}$ Term / last term

$a \rightarrow$ first term

$n \rightarrow$ number of terms in A.P

$d \rightarrow$ common difference

2. Write first four terms of the AP, when the first term a and the common difference d are given as follows:

(i) $a = 10, \quad d = 10$

(ii) $a = -2, \quad d = 0$

(iii) $a = 4, \quad d = -3$

(iv) $a = -1, \quad d = \frac{1}{2}$

(v) $a = -1.25, \quad d = -0.25$

3. For the following APs, write the first term and the common difference:

(i) $3, 1, -1, -3, \ldots$

(ii) $-5, -1, 3, 7, \ldots$

(iii) $\frac{1}{3}, \frac{5}{3}, \frac{9}{3}, \frac{13}{3}, \ldots$

(iv) $0.6, 1.7, 2.8, 3.9, \ldots$

4. Which of the following are APs ? If they form an AP, find the common difference d and write three more terms.

(i) $2, 4, 8, 16, \ldots$ ✗

(ii) $2, \frac{5}{2}, 3, \frac{7}{2}, \ldots$ ✓ $4, \frac{9}{2}, 5$ (½ ½ ½)

(iii) $-1.2, -3.2, -5.2, -7.2, \ldots$ $-9.2, -11.2, -13.2$

(iv) $-10, -6, -2, 2, \ldots$

(v) $3, 3+\sqrt{2}, 3+2\sqrt{2}, 3+3\sqrt{2}, \ldots$

(vi) $0.2, 0.22, 0.222, 0.2222, \ldots$

(vii) $0, -4, -8, -12, \ldots$

(viii) $-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \ldots$

(ix) $1, 3, 9, 27, \ldots$

(x) $a, 2a, 3a, 4a, \ldots$ 5a, 6a, 7a

(xi) $a, a^2, a^3, a^4, \ldots$

(xii) $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, \ldots$ $\sqrt{50}$ $\sqrt{72}$ $\sqrt{98}$

(xiii) $\sqrt{3}, \sqrt{6}, \sqrt{9}, \sqrt{12}, \ldots$

(xiv) $1^2, 3^2, 5^2, 7^2, \ldots$ 1 9 25

(xv) $1^2, 5^2, 7^2, 73, \ldots$ +24

$\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}, 5\sqrt{2}, 6\sqrt{2}, 7\sqrt{2}$

$\sqrt{2\times2\times2}$

$2\sqrt{2}$

EXERCISE 5.2

1. Fill in the blanks in the following table, given that a is the first term, d the common difference and a_n the nth term of the AP:

	a	d	n	a_n
(i)	7	3	8	28
(ii)	−18	2	10	0
(iii)	46	−3	18	−5
(iv)	−18.9	2.5	...	3.6
(v)	3.5	0	105	...

(i) $a_n = a + (n-1)d$

$a_8 = a + 7d$

$= 7 + 7(3)$

$= 7 + 21$

$= 28$

(ii) $a_n = a + (n-1)d$

$0 = -18 + 9d$

$18 = 9d \Rightarrow \boxed{d = 2}$

(iii) $a_n = a + (n-1)d$

$-5 = a + 17(-3)$

$-5 = a - 51$

$51 - 5 = a$

$\boxed{a = 46}$

EXERCISE 5.2

1. Fill in the blanks in the following table, given that a is the first term, d the common difference and a_n the nth term of the AP:

	a	d	n	a_n
(i)	7	3	8	28
(ii)	−18	2	10	0
(iii)	46	−3	18	−5
(iv)	−18.9	2.5	10	3.6
(v)	3.5	0	105	3.5

(iv) $a_n = a + (n-1)d$

$3.6 = -18.9 + (n-1)2.5$

$22.5 = (n-1)2.5$

$\frac{22.5}{2.5} = n-1$

$\Rightarrow 9 = n-1$

$10 = n$

$\frac{25}{2.5} = \frac{25}{25} \times 10$

2. Choose the correct choice in the following and justify :

(i) 30th term of the AP: 10, 7, 4, . . . , is

(A) 97 (B) 77 (C) –77 (D) – 87

HW (ii) 11th term of the AP: $-3, -\frac{1}{2}, 2, \ldots$, is

(A) 28 (B) 22 (C) –38 (D) $-48\frac{1}{2}$

3. In the following APs, find the missing terms in the boxes :

(i) 2, ☐, 26

(ii) ☐, 13, ☐, 3

(iii) 5, 6.5, 8, $9\frac{1}{2}$

(iv) – 4, ☐, ☐, ☐, ☐, 6

(v) ☐, 38, ☐, ☐, ☐, – 22

$a = 10, d = -3, a_{30}$

$a_n = a + (n-1)d$

$a_{30} = a + 29d$

$= 10 + 29(-3)$

$= 10 - 87$

$= -77$

(iii) $a = 5, a_n = 9.5$

$n = 4, d = ?$

$a_n = a + (n-1)d$

$9.5 = 5 + 3d$

$4.5 = 3d$

$\frac{4.5}{3} = d \Rightarrow d = 1.5$

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(A) 97 (B) 77 (C) –77 (D) – 87

HW (ii) 11th term of the AP: $-3, -\frac{1}{2}, 2, \ldots$, is

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3. In the following APs, find the missing terms in the boxes :

(i) 2, ☐, 26

(ii) ☐, 13, ☐, 3

(iii) 5, 6.5, 8, $9\frac{1}{2}$

(iv) – 4, ☐, ☐, ☐, ☐, 6

(v) 53, 38, 23, 8, -7, – 22

$a + d = 38$

$a + 5d = -22$

$-4d = 60$

$d = \frac{60}{-4} = -15$

$a - 15 = 38$

$a = 53$

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4. Which term of the AP : 3, 8, 13, 18, . . . ,is 78?

5. Find the number of terms in each of the following APs :

(i) 7, 13, 19, . . . , 205 (ii) 18, $15\frac{1}{2}$, 13, . . . , –47

6. Check whether – 150 is a term of the AP : 11, 8, 5, 2 . . .

7. Find the 31st term of an AP whose 11th term is 38 and the 16th term is 73.

$a = 3$, $d = 5$, $a_n = 78$, $n = ?$ (16)

→ Pair of linear equation.

(5) (ii) $a = 18$

$d = -2.5$

$a_n = -47$

$n = ?$

$a_{27} = -47$

$a_n = a + (n-1)d$

$-47 = 18 + (n-1)(-2.5)$

$-65 = (n-1)(-2.5)$

$\frac{+65}{+2.5} = n - 1$

$\frac{650}{25} = n - 1$

$26 = n - 1 \Rightarrow n = 27$

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8. An AP consists of 50 terms of which 3rd term is 12 and the last term is 106. Find the 29th term.

9. If the 3rd and the 9th terms of an AP are 4 and – 8 respectively, which term of this AP is zero? HW

10. The 17th term of an AP exceeds its 10th term by 7. Find the common difference.

11. Which term of the AP : 3, 15, 27, 39, . . . will be 132 more than its 54th term? HW

(8) $a_3 = 12$

$a_{50} = 106$

$a_{29} = ?$

$a_{29} = a + 28d$

$= 8 + 28(2)$

$= 8 + 56$

$= 64$

$\rightarrow a + 2d = 12$

$a + 49d = 106$

$-47d = -94$

$47d = 94$

$d = 2$

$a + 4 = 12$

$a = 8$

(10) $a_{17} - a_{10} = 7$

$(a + 16d) - (a + 9d) = 7$

$a + 16d - a - 9d = 7$

$7d = 7$

$d = 1$

12. Two APs have the same common difference. The difference between their 100th terms is 100, what is the difference between their 1000th terms?

13. How many three-digit numbers are divisible by 7?

14. How many multiples of 4 lie between 10 and 250?

15. For what value of n, are the nth terms of two APs: 63, 65, 67, . . . and 3, 10, 17, . . . equal?

16. Determine the AP whose third term is 16 and the 7th term exceeds the 5th term by 12.

(12)

AP1

$a = A_1$

$d = d$

AP2

$a = B_1$

$d = d$

100th term of AP1 $= a + 99d$

$= A_1 + 99d$

100th term $= B_1 + 99d$

$(A_1 + 99d) - (B_1 + 99d) = 100$

$A_1 + 99d - B_1 - 99d = 100$

$\boxed{A_1 - B_1 = 100}$

1000th term $= A_1 + 999d$

9000th term $= B_1 + 999d$

$(A_1 + 999d) - (B_1 + 999d)$

$A_1 + 999d - B_1 - 999d$

$A_1 - B_1$

$\boxed{= 100}$

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16. Determine the AP whose third term is 16 and the 7th term exceeds the 5th term by 12. } H.W.

(13)

$7\overline{)100}\,14$ $\quad -7$ $\quad 30$ $\quad -28$ $\quad 2$

$7\overline{)105}\,15$ $\quad -7$ $\quad 35$ $\quad -35$ $\quad 0$

$7\overline{)999}\,142$ $\quad -7\downarrow$ $\quad 29$ $\quad -28$ $\quad 19$ $\quad -14$ $\quad (5)$

105, 112, 119, ... 994

$a = 105$

$d = 7$

$a_n = 994$

$n = ?$

17. Find the 20th term from the last term of the AP : 3, 8, 13, . . ., 253.

18. The sum of the 4th and 8th terms of an AP is 24 and the sum of the 6th and 10th terms is 44. Find the first three terms of the AP.

19. Subba Rao started work in 1995 at an annual salary of ₹ 5000 and received an increment of ₹ 200 each year. In which year did his income reach ₹ 7000?

20. Ramkali saved ₹ 5 in the first week of a year and then increased her weekly savings by ₹ 1.75. If in the nth week, her weekly savings become ₹ 20.75, find n.

20/05/26

Sum of n terms of an AP

For an AP : $\underbrace{a}_{a_1}, \underbrace{a+d}_{a_2}, \underbrace{a+2d}_{a_3}, \ldots\ldots\ldots\ldots \underbrace{a+(n-1)d}_{a_n}$

$S_n = a_1 + a_2 + a_3 + \ldots\ldots a_n$

$$S_n = \frac{n}{2}\left[a + a_n\right]$$

$$\frac{n}{2}\left[a + a + (n-1)d\right]$$

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$$

$$S_n = \frac{n}{2}\left[a + a_n\right]$$

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$$

$1, 2, 3, \ldots\ldots\ldots\ldots 30$

$a = 1$
$d = 1$
$n = 30$

$$S_{30} = \frac{30}{2}\left[1 + 30\right]$$
$$= 15 \times 31$$
$$= \underline{465 \text{ Rs}}$$

EXERCISE 5.3

$a(b+c) = ab+c$

1. Find the sum of the following APs:

(i) 2, 7, 12, . . ., to 10 terms.

(ii) –37, –33, –29, . . ., to 12 terms.

(iii) 0.6, 1.7, 2.8, . . ., to 100 terms.

(iv) $\frac{1}{15}, \frac{1}{12}, \frac{1}{10}, \ldots$, to 11 terms.

$\frac{1}{12} - \frac{1}{15} = \frac{5-4}{60} = \frac{1}{60}$

(i) $a = 2$, $d = 5$, $n = 10$

$S_{10} = \frac{10}{2}[4 + 9(5)]$

$= 5[4+45]$

$= 5 \times 49$

$= 245$

(iv) $a = \frac{1}{15}$, $d = \frac{1}{60}$, $n = 11$

$\rightarrow S_n = \frac{n}{2}[2a + (n-1)d]$

$= \frac{11}{2}\left[2\left(\frac{1}{15}\right) + (11-1)\frac{1}{60}\right]$

$S_n = \frac{11}{2}\left[\frac{2}{15} + 10\left(\frac{1}{60}\right)\right]$

$= \frac{11}{2}\left[\frac{2}{15} + \frac{1}{6}\right]$

$= \frac{11}{2}\left[\frac{4+5}{30}\right] = \frac{11}{2} \times \frac{3}{\cancel{30}_{10}} = \frac{33}{20}$

2. Find the sums given below :

(i) $7+10\frac{1}{2}+14+\ldots+84$

(ii) $34+32+30+\ldots+10$

(iii) $-5+(-8)+(-11)+\ldots+(-230)$

3. In an AP:

(i) given $a = 5, d = 3, a_n = 50$, find n and S_n.

(ii) given $a = 7, a_{13} = 35$, find d and S_{13}.

(iii) given $a_{12} = 37, d = 3$, find a and S_{12}.

(iv) given $a_3 = 15, S_{10} = 125$, find d and a_{10}.

(v) given $d = 5, S_9 = 75$, find a and a_9.

(vi) given $a = 2, d = 8, S_n = 90$, find n and a_n.

(vii) given $a = 8, a_n = 62, S_n = 210$, find n and d.

(viii) given $a_n = 4, d = 2, S_n = -14$, find n and a.

(ix) given $a = 3, n = 8, S = 192$, find d.

(x) given $l = 28, S_9 = 144$, and there are total 9 terms. Find a.

(i) $a_n = a + (n-1)d$

$50 = 5 + (n-1)3$

$45 = 3(n-1)$

$\frac{45}{3} = n - 1$

$15 = n - 1$

$16 = n$

$S_{16} = \frac{16}{2}[5 + 50]$

$= 8 \times 55$

$= 440$

(ii)

4. How many terms of the AP : 9, 17, 25, . . . must be taken to give a sum of 636?

5. The first term of an AP is 5, the last term is 45 and the sum is 400. Find the number of terms and the common difference.

6. The first and the last terms of an AP are 17 and 350 respectively. If the common difference is 9, how many terms are there and what is their sum?

7. Find the sum of first 22 terms of an AP in which $d = 7$ and 22nd term is 149.

8. Find the sum of first 51 terms of an AP whose second and third terms are 14 and 18 respectively.

(4) $a = 9$

$d = 8$

$S_n = 636$

$n = ?$

$S_n = \frac{n}{2}[2a + (n-1)d]$

$636 = \frac{n}{2}[18 + (n-1)8]$

$636 = \frac{n}{2}[18 + 8n - 8]$

$636 = \frac{n}{2}(10 + 8n)$

$636 = n(5 + 4n)$

$636 = 5n + 4n^2$

$4n^2 + 5n - 636 = 0$

$a \quad b \quad c$

For a quadratic equation $ax^2 + bx + c = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$\Rightarrow \frac{-5 \pm \sqrt{25 - 4(4)(-636)}}{8}$

$= \frac{-5 \pm \sqrt{10201}}{8} = \frac{-5 \pm 101}{8} = \frac{96}{8}, \frac{-106}{8}$

$n = 12, \left(\frac{-106}{8}\right)$

4. How many terms of the AP : 9, 17, 25, . . . must be taken to give a sum of 636?

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6. The first and the last terms of an AP are 17 and 350 respectively. If the common difference is 9, how many terms are there and what is their sum?

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8. Find the sum of first 51 terms of an AP whose second and third terms are 14 and 18 respectively.

$a_n = 45$
$a = 5$
$S_n = 400$
$d = ?$
$n = 16$

$S_n = \frac{n}{2}[a + a_n]$

$400 = \frac{n}{2}[5 + 45]$

$400 = \frac{n}{2} \times 50^{25}$

$400 = 25n$

$n = \frac{400}{25} = 16$

$a_n = a + (n-1)d$

$45 = 5 + (16-1)d$

$45 = 5 + 15d$

$40 = 15d$

$8\frac{40}{15}{}^{3} = d$

$d = \frac{8}{3}$

4. How many terms of the AP : 9, 17, 25, . . . must be taken to give a sum of 636?

5. The first term of an AP is 5, the last term is 45 and the sum is 400. Find the number of terms and the common difference.

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7. Find the sum of first 22 terms of an AP in which $d = 7$ and 22nd term is 149.

8. Find the sum of first 51 terms of an AP whose second and third terms are 14 and 18 respectively.

$8 \Rightarrow a+d = 14 \quad (i)$

$a+2d = 18 \quad (ii)$

$-d = -4$

$d = 4$

By substituting value of d (i)

$a + d = 14$

$a + 4 = 14$

$a = 14 - 4$

$a = 10$

To find $\Rightarrow S_{51}$

$S_{51} = \frac{n}{2}(2a + 50d)$

$S_{51} = \frac{51}{2}(20 + 200)$

$= \frac{51}{2}(220)^{110}$

$S_{51} = 510$

9. If the sum of first 7 terms of an AP is 49 and that of 17 terms is 289, find the sum of first n terms.

10. Show that $a_1, a_2, \ldots, a_n, \ldots$ form an AP where a_n is defined as below :

(i) $a_n = 3 + 4n$ (ii) $a_n = 9 - 5n$

Also find the sum of the first 15 terms in each case.

11. If the sum of the first n terms of an AP is $4n - n^2$, what is the first term (that is S_1)? What is the sum of first two terms? What is the second term? Similarly, find the 3rd, the 10th and the nth terms.

12. Find the sum of the first 40 positive integers divisible by 6.

$S_7 = 49 \Rightarrow \frac{7}{2}(2a+(7-1)d) = 49$

$\Rightarrow \frac{7}{2}(2a+6d) = 49$

$\Rightarrow 7(a+3d) = 49$

$a+3d = 7$

$S_{17} = 289 \Rightarrow \frac{17}{2}(2a+16d) = 289$

$17(a+8d) = 289$

$a+8d = 17$ —

$a+3d = 7$

$5d = 10$

$d = 2$

$a+3(2) = 7$

$a = 1$

$S_n = ?$

$S_n = \frac{n}{2}(2a+(n-1)d)$

$= \frac{n}{2}(2+(n-1)2)$

$\frac{n}{2}(2+2n-2)$

$\frac{n}{2}(2n)$

$= n^2$

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13. Find the sum of the first 15 multiples of 8.

14. Find the sum of the odd numbers between 0 and 50.

15. A contract on construction job specifies a penalty for delay of completion beyond a certain date as follows: ₹ 200 for the first day, ₹ 250 for the second day, ₹ 300 for the third day, etc., the penalty for each succeeding day being ₹ 50 more than for the preceding day. How much money the contractor has to pay as penalty, if he has delayed the work by 30 days?

16. A sum of ₹ 700 is to be used to give seven cash prizes to students of a school for their overall academic performance. If each prize is ₹ 20 less than its preceding prize, find the value of each of the prizes.

17. In a school, students thought of planting trees in and around the school to reduce air pollution. It was decided that the number of trees, that each section of each class will plant, will be the same as the class, in which they are studying, e.g., a section of Class I will plant 1 tree, a section of Class II will plant 2 trees and so on till Class XII. There are three sections of each class. How many trees will be planted by the students?

18. A spiral is made up of successive semicircles, with centres alternately at A and B, starting with centre at A, of radii 0.5 cm, 1.0 cm, 1.5 cm, 2.0 cm, . . . as shown in Fig. 5.4. What is the total length of such a spiral made up of thirteen consecutive semicircles? (Take $\pi = \frac{22}{7}$)

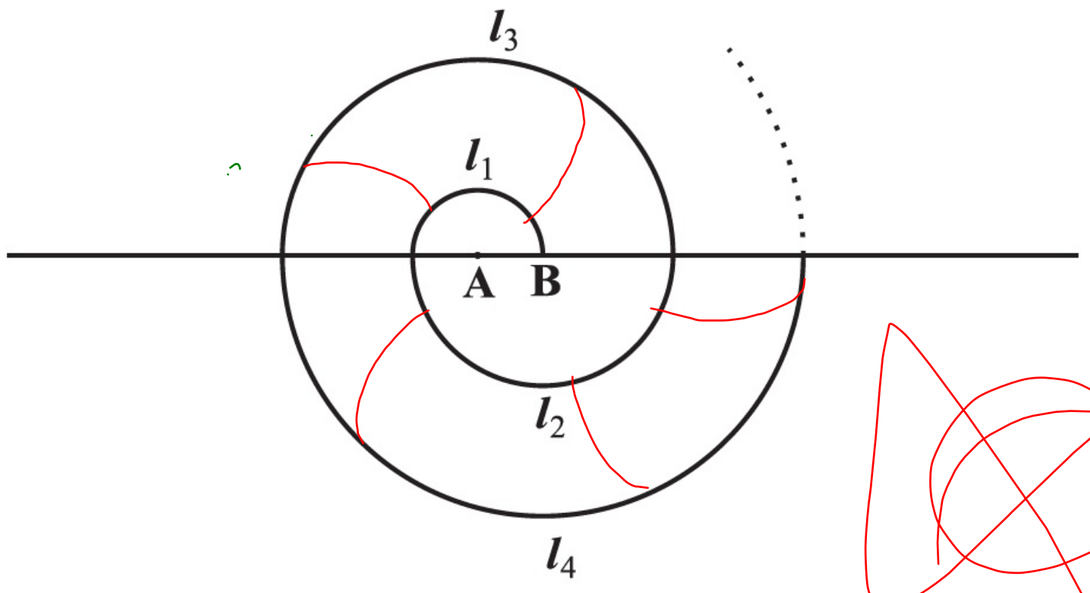


Fig. 5.4

[**Hint :** Length of successive semicircles is $l_1, l_2, l_3, l_4, \ldots$ with centres at A, B, A, B, . . ., respectively.]

19. 200 logs are stacked in the following manner: 20 logs in the bottom row, 19 in the next row, 18 in the row next to it and so on (see Fig. 5.5). In how many rows are the 200 logs placed and how many logs are in the top row?

41
×41
41
164×
1681

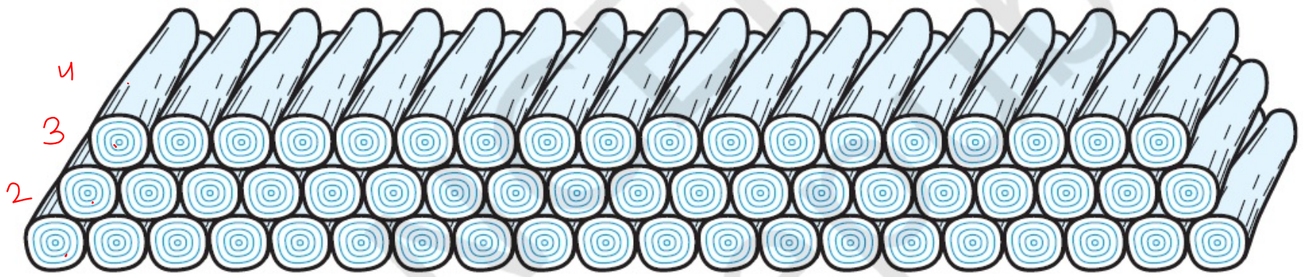


Fig. 5.5

$an^2+bn+c=0$

$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

$a=20$
$d=-1$
$S_n=200$
$n=?$
$a_n=$

$S_n=\frac{n}{2}[2a+(n-1)d]$

$200=\frac{n}{2}[40+(n-1)(-1)]$

$200=\frac{n}{2}[40-n+1]$

$200=\frac{n}{2}[41-n]$

$400=n(41-n)$

$400=41n-n^2$

$n^2-41n+400=0$

$n=\frac{41\pm\sqrt{1681-4(1)(400)}}{2}$

$=\frac{41\pm9}{2}=25, 16$

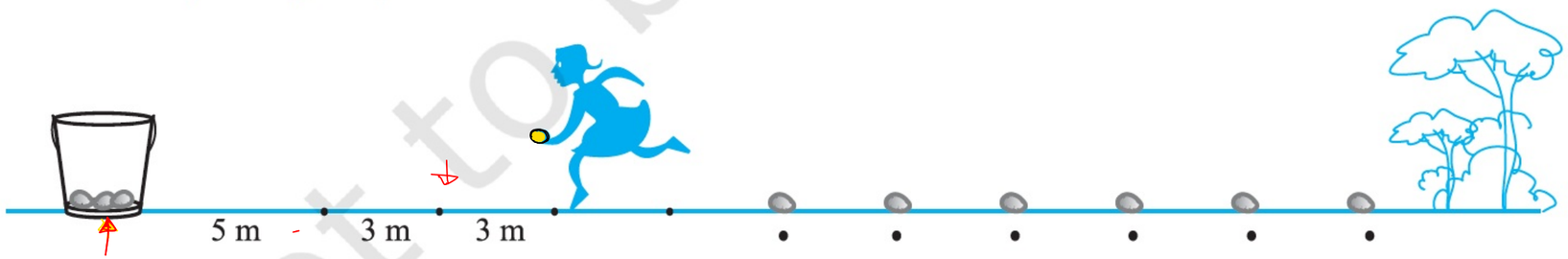
25 24 23 22 21th 20th 19th 18th 17th — 16th

−4 −3 −2 −1 0 1 2 3 4 5

0 0 00 000 0000 → 00000

$a_{16}=a+15d$
$=20+15(-1)$
$=5$

20. In a potato race, a bucket is placed at the starting point, which is 5 m from the first potato, and the other potatoes are placed 3 m apart in a straight line. There are ten potatoes in the line (see Fig. 5.6).



10m, 16m, 22m, ...

Fig. 5.6

A competitor starts from the bucket, picks up the nearest potato, runs back with it, drops it in the bucket, runs back to pick up the next potato, runs to the bucket to drop it in, and she continues in the same way until all the potatoes are in the bucket. What is the total distance the competitor has to run?

[**Hint :** To pick up the first potato and the second potato, the total distance (in metres) run by a competitor is $2 \times 5 + 2 \times (5 + 3)$]

$a = 10$ $\quad S_{10} = ?$
$d = 6$
$n = 10$

$S_n = \frac{n}{2}[a + a_n]$ or $\frac{n}{2}[2a + (n-1)d]$

$= \frac{10}{2}[20 + 9(6)]$

$= 5(20 + 54)$

$= 5(74) = 370\text{ m}$